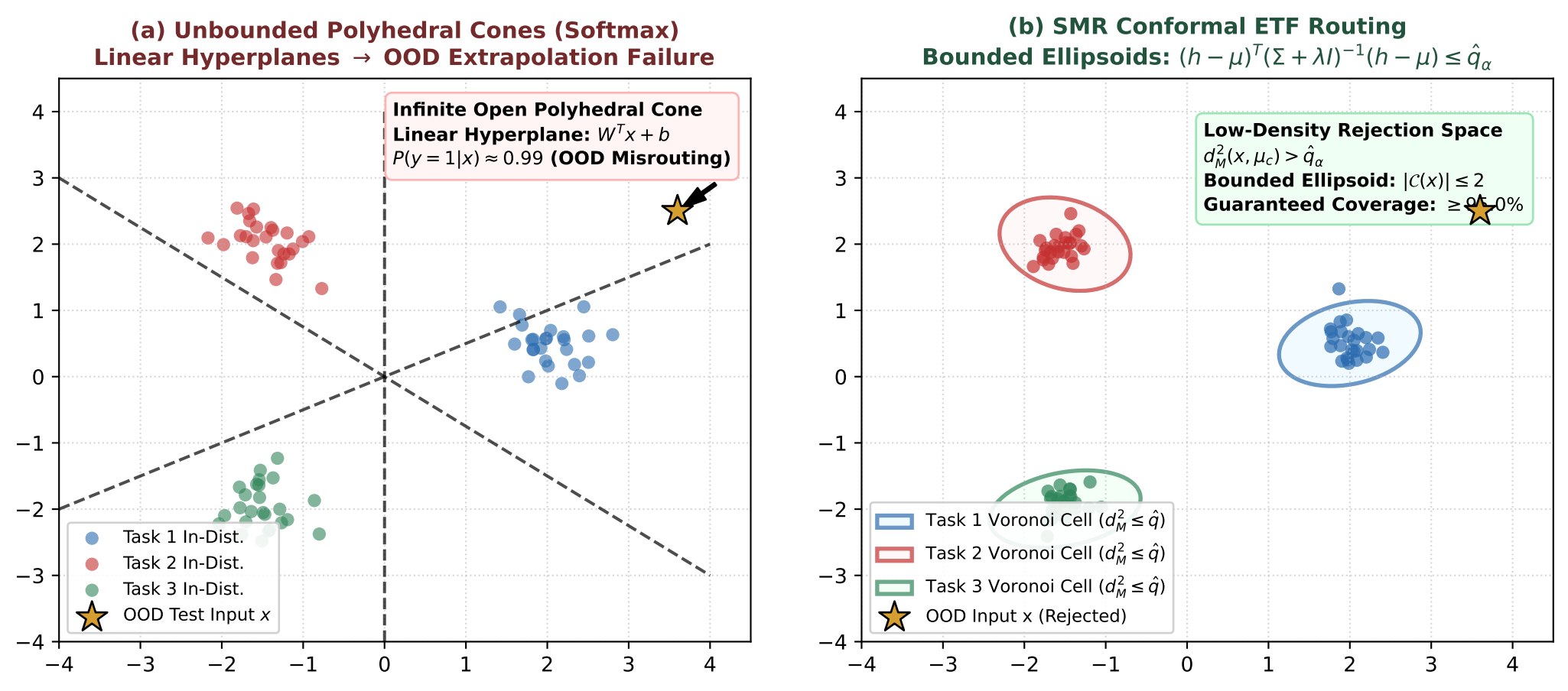
(a) Unbounded Polyhedral Cones (Softmax)
Linear Hyperplanes → OOD Extrapolation Failure
Infinite Open Polyhedral Cone
Linear Hyperplane: $W^T x + b$
$P(y = 1|x) \approx 0.99$ (OOD Misrouting)
Task 1 In-Dist.
Task 2 In-Dist.
Task 3 In-Dist.
OOD Test Input x
(b) SMR Conformal ETF Routing
Bounded Ellipsoids: $(h - \mu)^T(\Sigma + \lambda I)^{-1}(h - \mu) \leq \hat{q}_\alpha$
Low-Density Rejection Space
$d_M^2(x, \mu_c) > \hat{q}_\alpha$
Bounded Ellipsoid: $|\mathcal{C}(x)| \leq 2$
Guaranteed Coverage: $\geq 90\%$
Task 1 Voronoi Cell ($d_M^2 \leq \hat{q}$)
Task 2 Voronoi Cell ($d_M^2 \leq \hat{q}$)
Task 3 Voronoi Cell ($d_M^2 \leq \hat{q}$)
OOD Input x (Rejected)